

Machine Learning

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions or decisions.

Supervised learning uses labelled data for tasks such as classification and regression. Unsupervised learning discovers patterns in unlabelled data using techniques such as clustering.

A machine learning workflow includes data collection, preprocessing, feature engineering, model training, validation, evaluation, and deployment.

Common evaluation metrics include accuracy, precision, recall, F1-score, mean squared error, and mean absolute error.

Machine learning is used in cybersecurity, healthcare, finance, recommendation systems, computer vision, and natural language processing.